

DESi epistemic-state ablation — consolidated report (Phases 2–5)

A falsification-oriented study of whether a DESi epistemic-state slice helps an LLM, and *why* — vs full chat, vs plausible-wrong / neutral / contradictory states, vs lexical and neural retrieval, with metadata on/off, and with the state hand-curated vs auto-constructed. Real runs (OpenRouter), temperature 0, fixed seed, 2–3 reps; 4 models (Sonnet-4.5, GPT-4o, Llama-3.3-70B, Granite-4.1-8B). The recall evaluator is paraphrase- and negation-blind (content-token Jaccard ≥ 0.25), so only RELATIVE comparisons are meant; no prompt was tuned to make any condition win. **Bottom line up front:** correct, auto-constructible state selection is load-bearing and is mainly a TOKEN-EFFICIENCY + long-document-robustness win over retrieval — not a general capability gain, and metadata governance is not established.

1 · The condition ladder, recall across regimes

regime (model)	A	B	B_auto	C	F	G	R1	R2n	E
core short (sonnet)	0.712	0.881	—	0.019	0.019	0.019	0.462	—	0.879
lifecycle short (sonnet)	0.938	1.0	0.95	—	—	—	0.537	0.85	0.988
lifecycle short (gpt-4o)	0.85	0.988	0.95	—	—	—	0.488	0.713	1.0
lifecycle short (llama)	0.725	1.0	0.875	—	—	—	0.513	0.725	0.975
lifecycle LONG ~18k (sonnet)	0.88	1.0	0.96	—	—	—	0.52	0.08	1.0
lifecycle LONG ~18k (granite)	0.76	0.96	0.88	—	—	—	0.44	0.1	0.94

Blank = condition not run in that regime. C wrong-slice / F empty / G neutral collapse everywhere; B and B_auto stay near 1.0.

2 · The five questions

- **Q1 Is correct state SELECTION load-bearing? YES (strong).** Wrong-slice C, empty F and neutral G all collapse to ~0 recall while B $\approx$ 1.0 — across every regime and all 4 models. A plausible-but-wrong state is no better than no state, and worse on degeneration.
- **Q2 Does wrong state POISON? YES — visible in degeneration, not recall.** On recall $F \approx G \approx C$ (~0); the degeneration metrics separate them: C adopts the wrong case's claims (invalid-reuse $\approx$ 9–15, bad-framing 1.0), G is pulled into irrelevant content ($\approx$ 3–8), H (contradiction) loops and reuses contradictions. Injected claims PERSIST and even RELAPSE across neutral double-check probes (C: dropped when probed, then reverted). The model is CONFIDENTLY wrong (self-reported confidence 93–100 while recall ~0).
- **Q3 Does the metadata GOVERNANCE help? NOT established.** Budget-matched B–E $\approx$ 0 in every phase and every model (the category typing / IDs add no measurable recall over the same texts at the same budget). The value is the SELECTION, not the typing.
- **Q4 Does DESi beat RETRIEVAL? Yes, but the honest size depends on the retriever and the document length (see §3).**
- **Q5 Can the state be AUTO-CONSTRUCTED? YES.** One LLM extraction pass that resolves supersession yields B_auto $\approx$ B (B–B_auto $\approx$ +0.04..+0.13, within noise), worst single case 0.80, on all models — and it survives long documents (B_auto 0.88–0.96 at ~18k tokens). So the Phase-3 'curated-GT' caveat is substantially addressed: the value is constructible, even by a small model (Granite).

3 · The DESi-vs-retrieval arc (the most nuanced result)

retriever / regime	B – retrieval (recall)	reading
lexical BM25/TF-IDF, short docs (P3)	$\approx$ +0.70	lexical retrieval overstated DESi's edge
neural bge-small, budget-matched, short (P4)	$\approx$ +0.15 .. +0.28	a good dense retriever closes most of the gap

retriever / regime	B – retrieval (recall)	reading
neural, OPTIMAL $k \approx$ all chunks, short (k-sweep)	$\approx +0.05$	at unlimited budget retrieval $\approx$ DESi
budget-matched, LONG $\sim 18k$ docs (P5)	BM25 $+0.5 \cdot$ neural $+0.9$	retrieval collapses; DESi holds at $\sim 2\%$ of tokens

The k-sweep (Sonnet, 8 short cases) is the key to reading this: neural recall ran 0.15 ($k=1$) $\rightarrow$ 0.83 ($k \approx 13$, $1 \times B$ budget) $\rightarrow$ 0.95 ($k \approx 18 \approx$ all chunks). So on SHORT chats retrieval matches DESi *if given enough k*, which is cheap there — meaning DESi's short-doc edge is mostly TOKEN EFFICIENCY (same recall, ~ 13 chunks vs a 372-token slice), not capability. On LONG documents 'all chunks' does not fit the budget: at 1% of ~ 990 chunks the small embedder is fooled by generically-relevant filler (it retrieves 'how is X set up' over the actual decision turns; BM25's keyword match is more robust), so retrieval collapses while DESi/B_auto stay near 1.0. *Honest caveat*: the neural collapse is for a SMALL embedder (bge-small 384d) + a generic-task query; a stronger embedder would rank needles better, so $B-R1 \approx +0.5$ is the more representative long-doc gap.

4 • Token efficiency (the clearest practical win)

On the $\sim 18k$ -token documents, B (DESi) scored 1.0 / 0.96 (Sonnet / Granite) on **~ 372 input tokens**, while the full chat A scored 0.88 / 0.76 on **~ 18342 tokens** ($\sim 49\times$ more) — i.e. DESi gives **better-than-full-context recall at $\sim 1/49$ of the cost**, and B_auto nearly matches it. This is the most robust, least-caveated finding.

5 • Consolidated conclusions

Supported by the data:

- Correct, AUTO-CONSTRUCTIBLE epistemic-state selection is load-bearing.
- Plausible-wrong / neutral / contradictory structure is actively toxic, sticky (persists & relapses) and produces confidently-wrong output — detectable only via degeneration metrics.
- DESi's advantage over retrieval is TOKEN EFFICIENCY (full-context-or-better recall at $\sim 2\%$ of tokens) + ROBUSTNESS on long documents where budget retrieval cannot fit/find the needles.
- Confidently-reporting-stale is primarily a RETRIEVAL pathology; the resolved state removes it.

NOT supported / explicitly not claimed:

- 'DESi makes the model smarter' / 'DESi wins generally' — no general capability gain; at unlimited budget a good neural retriever $\approx$ DESi on short tasks.
- Metadata governance (typing/IDs) — $B \approx E$ everywhere; not established.
- A large absolute-capability gap vs strong retrieval — the lexical ~ 0.7 was an artefact.

Open: a held-out extractor model for B_auto (avoid self-circularity); a stronger neural retriever for the long-doc gap; harder multi-supersession chains; larger N (15–20+) to tighten the borderline short-doc $B-R2n$ CI.

6 • Limitations & reproduce

Small N per regime (2–8 cases); paraphrase/negation-blind evaluator (relative only; recall is unreliable for the contradiction condition H); B_auto extractor = same model family as the summariser; long-doc filler is synthetic; self-confidence is self-reported, not logprobs. All conditions/metrics are unit-tested; runs are temp-0 + seeded; results store metrics only (no raw model text, no API key). Reproduce: `export OPENROUTER_API_KEY=...; python ab_evidence/ablation_run2.py; build PDFs via ab_evidence/build_phase{2..5}_pdf.py`. Total spend across all phases $\approx$ a few USD.